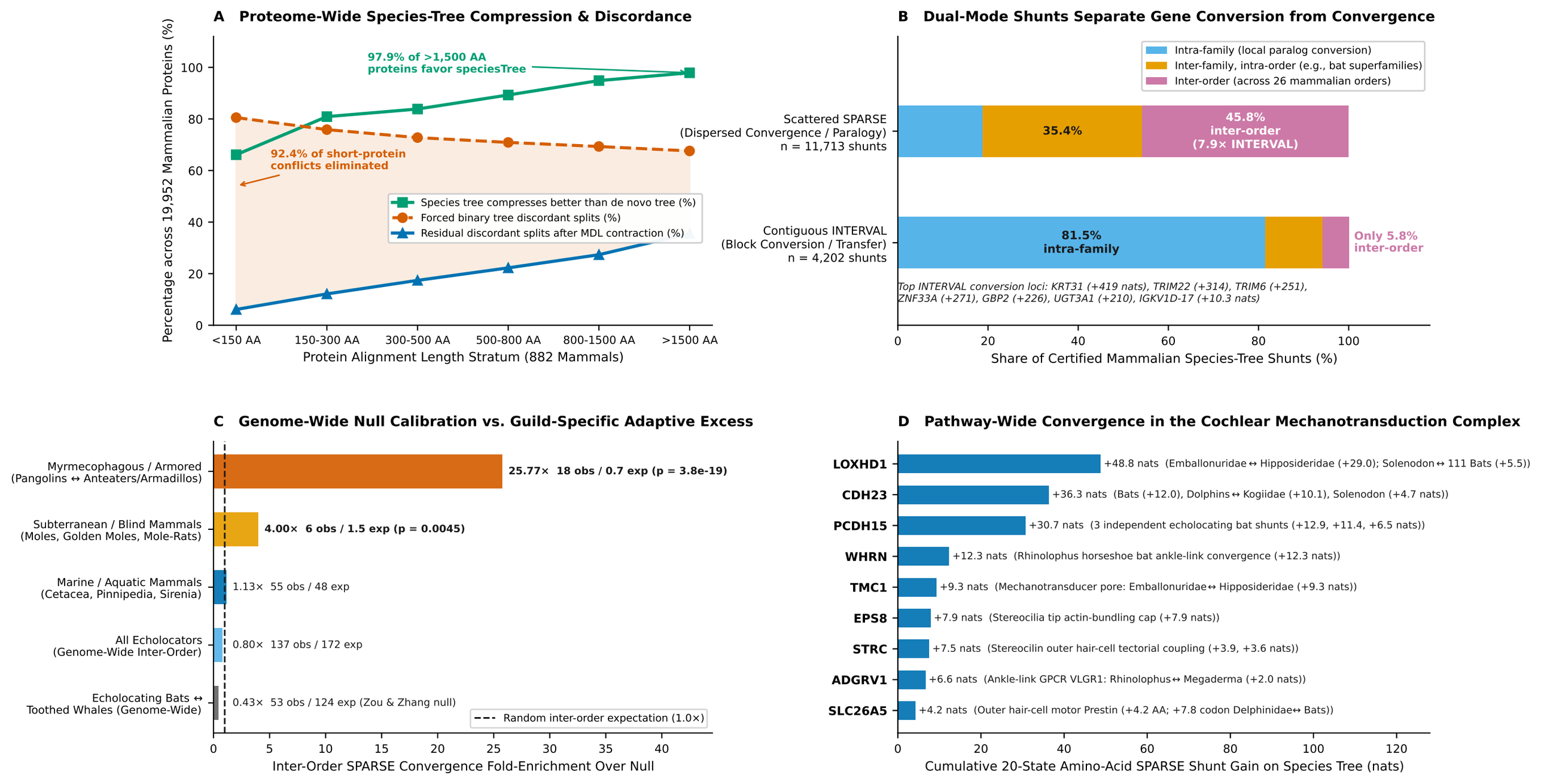

A Proteome-Wide Species-Tree Compression & Discordance
97.9% of >1,500 AA proteins favor speciesTree
92.4% of short-protein conflicts eliminated
Species tree compresses better than de novo tree (%)
Forced binary tree discordant splits (%)
Residual discordant splits after MDL contraction (%)
Percentage across 19,952 Mammalian Proteins (%)
<150 AA
150-300 AA
300-500 AA
500-800 AA
800-1500 AA
>1500 AA
Protein Alignment Length Stratum (882 Mammals)
B Dual-Mode Shunts Separate Gene Conversion from Convergence
Intra-family (local paralog conversion)
Inter-family, intra-order (e.g., bat superfamilies)
Inter-order (across 26 mammalian orders)
Scattered SPARSE
(Dispersed Convergence / Paralogy)
n = 11,713 shunts
35.4%
45.8%
inter-order
(7.9× INTERVAL)
Contiguous INTERVAL
(Block Conversion / Transfer)
n = 4,202 shunts
81.5%
intra-family
Only 5.8%
inter-order
Top INTERVAL conversion loci: KRT31 (+419 nats), TRIM22 (+314), TRIM6 (+251),
ZNF33A (+271), GBP2 (+226), UGT3A1 (+210), IGKV1D-17 (+10.3 nats)
Share of Certified Mammalian Species-Tree Shunts (%)
C Genome-Wide Null Calibration vs. Guild-Specific Adaptive Excess
Myrmecophagous / Armored
(Pangolins ↔ Anteaters/Armadillos)
25.77× 18 obs / 0.7 exp (p = 3.8e-19)
Subterranean / Blind Mammals
(Moles, Golden Moles, Mole-Rats)
4.00× 6 obs / 1.5 exp (p = 0.0045)
Marine / Aquatic Mammals
(Cetacea, Pinnipedia, Sirenia)
1.13× 55 obs / 48 exp
All Echolocators
(Genome-Wide Inter-Order)
0.80× 137 obs / 172 exp
Echolocating Bats ↔
Toothed Whales (Genome-Wide)
0.43× 53 obs / 124 exp (Zou & Zhang null)
Random inter-order expectation (1.0×)
Inter-Order SPARSE Convergence Fold-Enrichment Over Null
D Pathway-Wide Convergence in the Cochlear Mechanotransduction Complex
LOXHD1 +48.8 nats (Emballonuridae ↔ Hipposideridae (+29.0); Solenodon ↔ 111 Bats (+5.5))
CDH23 +36.3 nats (Bats (+12.0), Dolphins ↔ Kogiidae (+10.1), Solenodon (+4.7 nats))
PCDH15 +30.7 nats (3 independent echolocating bat shunts (+12.9, +11.4, +6.5 nats))
WHRN +12.3 nats (Rhinolophus horseshoe bat ankle-link convergence (+12.3 nats))
TMC1 +9.3 nats (Mechanotransducer pore: Emballonuridae ↔ Hipposideridae (+9.3 nats))
EPS8 +7.9 nats (Stereocilia tip actin-bundling cap (+7.9 nats))
STRC +7.5 nats (Stereocilin outer hair-cell tectorial coupling (+3.9, +3.6 nats))
ADGRV1 +6.6 nats (Ankle-link GPCR VLGR1: Rhinolophus ↔ Megaderma (+2.0 nats))
SLC26A5 +4.2 nats (Outer hair-cell motor Prestin (+4.2 AA; +7.8 codon Delphinidae ↔ Bats))
Cumulative 20-State Amino-Acid SPARSE Shunt Gain on Species Tree (nats)